Practice Final A

MTH 207 (Fall 2026) — J. Baggett

This is practice, not a preview. The real exam will **not** be these problems with the numbers changed. Expect different functions, and expect at least one technique this sheet happens not to show. Everything on the review list is fair game whether or not it turns up here — use this to find out what you cannot yet do, not to predict what will be asked.

This sheet is also deliberately **longer** than the real exam (287 points against 200) so that it covers more ground, so do not use it to judge timing. The real exam is 120 minutes.

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
  - Explain your reasons and justify your steps.
  - Demonstrate that you understand the *process*.
  - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.
- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3pts ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) If $f'(2) = 0$, then f has a highest or lowest value at $x = 2$.

Solution:

FALSE. $f(x) = x^3$ has $f'(0) = 0$ with neither. A horizontal tangent does not have to be a peak or a valley.

(b) A secant line through two points on the graph of f can be horizontal even when f is never constant.

Solution:

TRUE. It needs only $f(a) = f(b)$. On $f(x) = (x-2)^2$ the secant through $x = 1$ and $x = 3$ is horizontal, and the function is constant nowhere.

(c) $\dfrac{d}{dx}\sin\left(x^2\right) = \cos(2x)$.

Solution:

FALSE. The outer derivative is evaluated *at the inner function* and then multiplied by its derivative: $\cos(x^2) \cdot 2x$. Differentiating the inside in place is not the Chain Rule.

(d) If $\lim\limits_{x\to a} f(x) = 0$ and $|g(x)| \le 1$ for all x, then $\lim\limits_{x\to a} f(x)g(x) = 0$.

Solution:

TRUE. Squeeze Theorem: $-|f| \le fg \le |f|$ and both bounds go to 0. Note $\lim g$ need not exist.

(e) If f has an absolute minimum value at c, then $f'(c) = 0$.

Solution:

FALSE. c could be an endpoint, or f' could fail to exist there – $f(x) = |x|$ at 0.

(18pts) **2.** Find a point on the curve $x = 4 - y^2$ which is closest to the origin.

Solution:
Minimize the *square* of the distance — same minimizer, no square root to differentiate. A point on the curve is $(4-y^2,\, y)$, so

$$D = x^2 + y^2 = (4-y^2)^2 + y^2$$

Let $u = y^2$ $(u \geq 0)$: $D = (4-u)^2 + u = u^2 - 7u + 16$.

$$\frac{dD}{du} = 2u - 7 = 0 \quad \Longrightarrow \quad u = \frac{7}{2}$$

$\dfrac{d^2D}{du^2} = 2 > 0$, so this is a minimum. Then $y^2 = \dfrac{7}{2}$ and $x = 4 - \dfrac{7}{2} = \dfrac{1}{2}$.

Points: $\left(\frac{1}{2},\, \frac{\sqrt{14}}{2}\right)$ and $\left(\frac{1}{2},\, -\frac{\sqrt{14}}{2}\right)$

(Both, by symmetry.) The distance is $\sqrt{15}/2 \approx 1.94$.

(12pts) **3.** For the equation $y^3 - xy = -6$ find the equation of the tangent line to the curve at the point $(1, -2)$.

Solution:
First check the point is on the curve: $(-2)^3 - (1)(-2) = -8 + 2 = -6$. ✓
Differentiate implicitly (the xy term needs the Product Rule):

$$3y^2\frac{dy}{dx} - \left(y + x\frac{dy}{dx}\right) = 0 \quad \Longrightarrow \quad \frac{dy}{dx}\left(3y^2 - x\right) = y \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{y}{3y^2 - x}$$

At $(1, -2)$:

$$\frac{dy}{dx} = \frac{-2}{3(4) - 1} = -\frac{2}{11}$$

Tangent line: $y + 2 = -\dfrac{2}{11}(x - 1)$ or $y = -\dfrac{2}{11}x - \dfrac{20}{11}$

(4pts) **4.** State the definition of "f is continuous at a".

Solution:
f is **continuous at** a means

$$\lim_{x \to a} f(x) = f(a)$$

That single equation quietly requires three things, and naming them is worth the points:

1. $f(a)$ is defined;
2. $\lim\limits_{x \to a} f(x)$ exists;
3. the two are equal.

(3pts) **5.** Give an example of a function f and an x-value a where f is continuous at a but not differentiable at a.

Solution:
$f(x) = |x|$ at $a = 0$.
Continuous: $\lim_{x \to 0} |x| = 0 = f(0)$.

Not differentiable: the difference quotient is $\dfrac{|h|}{h}$, which is $+1$ for $h > 0$ and -1 for $h < 0$. The one-sided limits disagree, so $f'(0)$ does not exist — the graph has a **corner**.
(Other correct answers: $f(x) = \sqrt[3]{x}$ at 0, which has a vertical tangent; or $f(x) = |x - 2|$ at 2.)

(10pts) **6.** Consider the region bounded by $y = 2^x$, $y = 3^x$ and $x = 5$. What is the area of the region?

Solution:

The two exponentials meet where $2^x = 3^x$, i.e. at $x = 0$ (both equal 1). For $x > 0$, $3^x > 2^x$, so $y = 3^x$ is the upper curve on $[0, 5]$.

$$A = \int_0^5 (3^x - 2^x)\,dx = \left[\frac{3^x}{\ln 3} - \frac{2^x}{\ln 2}\right]_0^5$$

$$= \left(\frac{243}{\ln 3} - \frac{32}{\ln 2}\right) - \left(\frac{1}{\ln 3} - \frac{1}{\ln 2}\right) = \frac{\mathbf{242}}{\ln \mathbf{3}} - \frac{\mathbf{31}}{\ln \mathbf{2}} \approx 175.6$$

Recall $\int b^x dx = \dfrac{b^x}{\ln b} + C$.

(6pts ea.) **7.** Evaluate the following limits. As a reminder, justify your steps! Simplify your answer as much as possible. (If the limit does not exist, state why.)

(a) $\displaystyle\lim_{x\to\infty} \frac{\sqrt{9x^6 - x}}{x^3 + 1}$

(b) $\displaystyle\lim_{x\to 0} \frac{\cos x - 1}{\sin x}$

(c) $\displaystyle\lim_{x\to\infty} \left(1 - \frac{1}{x}\right)^x$

(7^{pts} ea.) **8.** Find the derivatives (with respect to x) of the following functions:

(a) $f(x) = \dfrac{\sin 3x}{\tan^{-1} x}$

(b) $f(x) = \displaystyle\int_3^x \sin(t^9)\, dt$

(c) $f(x) = (\cos x)^{\log_3 x}$

(d) $f(x) = \displaystyle\int_x^2 \cos(t^3)\, dt$

(8 pts) **9.** Given a fixed number $b > 0$, derive the derivative rule for $y = b^x$. (You may use the derivative rule for e^x.)

Solution:

Rewrite the base as a power of e. Since $b = e^{\ln b}$,

$$y = b^x = \left(e^{\ln b}\right)^x = e^{x \ln b}$$

Now differentiate with the Chain Rule, treating $\ln b$ as the constant it is:

$$\frac{dy}{dx} = e^{x \ln b} \cdot \frac{d}{dx}(x \ln b) = e^{x \ln b} \cdot \ln b$$

and $e^{x \ln b} = b^x$, so

$$\boxed{\frac{d}{dx}(b^x) = b^x \ln b}$$

Sanity check: $b = e$ gives $\ln e = 1$ and $(e^x)' = e^x$. ✓

(7^{pts} ea.) **10.** Evaluate the following indefinite integrals. Simplify your answer as much as possible.

(a) $\displaystyle\int u(u-1)^8\,du$

(b) $\displaystyle\int \frac{5-3x}{1+x^2}\,dx$

(c) $\displaystyle\int \frac{-1}{2^{\sin^{-1}x}\sqrt{1-x^2}}\,dx$

(8^{pts} ea.) **11.** Evaluate the following definite integrals. Simplify your answer as much as possible.

(a) $\displaystyle\int_1^7 |x-3|\,dx$

(b) $\displaystyle\int_{-1}^{\frac{1}{\sqrt{3}}} \frac{dx}{4x^2+4}$

(c) $\displaystyle\int_0^{81\pi^2} \frac{\sin(\sqrt{x})}{\sqrt{x}}\,dx$

(10 pts) **12.** Choose ONE integral on this page and evaluate it. Clearly mark which integral is your choice! (Suggestion: do your scratch work on the back page or ask for extra paper.)

$$\int \frac{e^{5x}}{e^{10x}+1}\,dx \qquad \int \frac{\sec x + \tan x}{\cos x}\,dx \qquad \int_{-1/3}^{\sqrt{6}/6} \frac{1}{\sqrt{2-9x^2}}\,dx \qquad \int \frac{x}{x^4+2x^2+2}\,dx$$

Solution:

Choosing the first integral. Note $e^{10x} = \left(e^{5x}\right)^2$, so the denominator is u^2+1 with $u = e^{5x}$.

Let $u = e^{5x}$, so $du = 5e^{5x}\,dx$, i.e. $e^{5x}dx = \dfrac{du}{5}$:

$$\int \frac{e^{5x}}{e^{10x}+1}\,dx = \int \frac{1}{u^2+1}\cdot\frac{du}{5} = \frac{1}{5}\arctan u + C = \mathbf{\frac{1}{5}}\arctan\left(\mathbf{e^{5x}}\right) + \mathbf{C}$$

(10pts) **13.** Use the definition of the derivative to compute $f'(x)$ if $f(x) = x^2 - 3x$.

Solution:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\left[(x+h)^2 - 3(x+h)\right] - \left[x^2 - 3x\right]}{h}$$
$$= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - 3x - 3h - x^2 + 3x}{h} = \lim_{h\to 0} \frac{2xh + h^2 - 3h}{h}$$
$$= \lim_{h\to 0} \frac{h\,(2x + h - 3)}{h} = \lim_{h\to 0} (2x + h - 3) = \mathbf{2x - 3}$$

Keep lim on every line until the substitution at the end, and note both brackets when subtracting $f(x)$ — the dropped minus sign is the usual error.

(15pts) **14.** Determine $\int_1^5 (3x - 2)\,dx$ as a limit of Riemann sums. (You may use any sampling points you like, but say which you have chosen.)

Solution:

Take n equal subintervals of $[1, 5]$, so $\Delta x = \dfrac{5-1}{n} = \dfrac{4}{n}$, and use **right** endpoints $x_i = 1 + i\Delta x = 1 + \dfrac{4i}{n}$.

$$f(x_i) = 3\left(1 + \frac{4i}{n}\right) - 2 = 1 + \frac{12i}{n}$$

$$\int_1^5 (3x - 2)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} \left(1 + \frac{12i}{n}\right)\frac{4}{n} = \lim_{n\to\infty} \left[\frac{4}{n}\sum_{i=1}^{n} 1 + \frac{48}{n^2}\sum_{i=1}^{n} i\right]$$

Using $\sum 1 = n$ and $\sum i = \dfrac{n(n+1)}{2}$,

$$= \lim_{n\to\infty} \left[4 + \frac{48}{n^2}\cdot\frac{n(n+1)}{2}\right] = \lim_{n\to\infty} \left[4 + 24\left(1 + \frac{1}{n}\right)\right] = 4 + 24 = \mathbf{28}$$

Check: $\left[\dfrac{3}{2}x^2 - 2x\right]_1^5 = (37.5 - 10) - (1.5 - 2) = 28$. ✓

(5pts) **15.** Given a function $f(x)$, explain what $\int_a^b f(x)\,dx$ represents in terms of the graph of f.

Solution:

$\int_a^b f(x)\,dx$ is the **net signed area** between the graph of f and the x-axis, from $x = a$ to $x = b$.

"Signed" is the whole point: area where $f(x) > 0$ counts **positively**, area where $f(x) < 0$ counts **negatively**, and the integral is the difference. So it can be positive, negative or zero — unlike an area, which never is.

(18pts) **16.** A spherical balloon is blown up using an air supply which operates at 2 cm^3/sec. How fast is the radius increasing when the radius is 5 cm?

Solution:

Relation: $V = \frac{4}{3}\pi r^3$, with V and r both functions of time.

Differentiate with respect to t (Chain Rule on the right):

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

Substitute the given values *after* differentiating — $\frac{dV}{dt} = 2\ \text{cm}^3/\text{sec}$ and $r = 5$ cm:

$$2 = 4\pi(5)^2 \frac{dr}{dt} = 100\pi \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{\mathbf{1}}{\mathbf{50}\pi}\ \text{cm/sec} \approx 0.0064\ \text{cm/sec}$$

(8 pts) **17.** Show that the equation $2x + \cos x = 0$ has exactly one root.

Solution:
Let $f(x) = 2x + \cos x$.
At least one root. f is continuous everywhere. And

$$f(-1) = -2 + \cos(-1) \approx -2 + 0.54 = -1.46 < 0, \qquad f(0) = 0 + 1 = 1 > 0$$

so by the **Intermediate Value Theorem** there is a root in $(-1, 0)$.
At most one root. $f'(x) = 2 - \sin x$. Since $-1 \le \sin x \le 1$, we have $f'(x) \ge 2 - 1 = 1 > 0$ for every x, so f is **strictly increasing** and can cross zero only once.
Hence **exactly one root**.

(2 pts) **18.** Consider $f(x) = \sqrt{x-2}$ on $[6, 18]$. According to the Mean Value Theorem there is a number c in $(6, 18)$ so that $f'(c) =$ _____. Compute and simplify the number to fill in the blank.

Solution:
The Mean Value Theorem says $f'(c)$ equals the average rate of change across the interval:

$$f'(c) = \frac{f(18) - f(6)}{18 - 6} = \frac{\sqrt{16} - \sqrt{4}}{12} = \frac{4 - 2}{12} = \mathbf{\frac{1}{6}}$$

(No need to find c itself — the blank asks only for the value of $f'(c)$.)

(5 pts) **19.** What is the linearization of $f(x) = x^3$ at 5?

Solution:
$L(x) = f(a) + f'(a)(x - a)$ with $a = 5$. Here $f(5) = 125$ and $f'(x) = 3x^2$ so $f'(5) = 75$:

$$L(x) = 125 + 75(x - 5) = \mathbf{75x - 250}$$

(8 pts) **20.** Find the local minimum(s) of $f(x) = x^4 + 4x^3 + 4x^2 - 1$. Find the local maximum(s).

Solution:

$$f'(x) = 4x^3 + 12x^2 + 8x = 4x\left(x^2 + 3x + 2\right) = 4x(x+1)(x+2)$$

so the critical numbers are $x = -2, -1, 0$.
Sign of f' across them (three linear factors):

	$x < -2$	$-2 < x < -1$	$-1 < x < 0$	$x > 0$
f'	$-$	$+$	$-$	$+$

By the First Derivative Test, f' changes $- \to +$ at $x = -2$ and at $x = 0$ (**local minima**) and $+ \to -$ at $x = -1$ (**local maximum**).

$$f(-2) = -1, \qquad f(-1) = 0, \qquad f(0) = -1$$

Local minima at $(-2, -1)$ and $(0, -1)$; local maximum at $(-1, 0)$.

(10^{pts}) **21.** Use the information given to sketch a graph of a function f

- f is continuous except at $x = -1$
- $f(-3) = -3, f(0) = 0, f(1) = 1, f(2) = 8/9$
- $f'(x) > 0$ on $(-1, 1)$ and $f'(x) < 0$ on $(-\infty, -1) \cup (1, \infty)$
- $f''(x) > 0$ on $(2, \infty)$ and $f''(x) < 0$ on $(-\infty, -1) \cup (-1, 2)$

Solution:
Read each clue as a feature of the picture:

- **Break at** $x = -1$: the graph is in two pieces, split at -1.
- $f' > 0$ **on** $(-1, 1)$**,** $f' < 0$ **on** $(-\infty, -1)$ **and** $(1, \infty)$: falling to the left of -1; rising from -1 up to 1; falling after 1. So there is a **local maximum at** $x = 1$, height $f(1) = 1$.
- $f'' < 0$ **on** $(-\infty, -1)$ **and** $(-1, 2)$**,** $f'' > 0$ **on** $(2, \infty)$: concave down everywhere left of 2, concave up after, so an **inflection point at** $x = 2$, height $\frac{8}{9}$.

A correct sketch passes through $(-3, -3)$, $(0, 0)$, $(1, 1)$ and $\left(2, \frac{8}{9}\right)$; falls and is concave down up to the break at -1; rises after the break, concave down, topping out at $(1, 1)$; then falls, changing from concave down to concave up at $x = 2$.

(3^{pts} ea.) **22.** An object moves along a horizontal coordinate line in such a way that its position at time t is given by $s(t) = t^3 - 11t^2 + 24t + 100$. Here s is measured in feet and t in seconds.

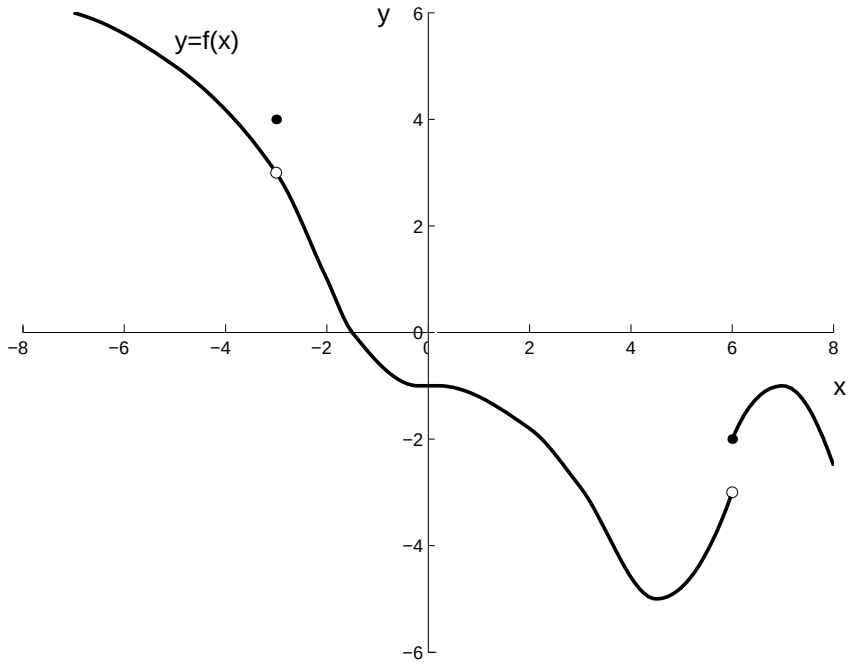
(a) What is the velocity of the object after 1 second?

(b) When is the object moving left?

(c) When is the acceleration positive?

(16 pts) **23.** From the graph of f, approximate:

Where $f(x) = 0$
Where $f'(x) = 0$
Where f is **not** continuous
Where $f'(x)$ is negative
Where f is **not** differentiable
Inflection point(s) of f
$\lim_{x \to -2} f(x)$
$\lim_{x \to 6^-} f(x)$



Solution:

Reading the graph (answers are approximations, as asked):

- $f(x) = 0$**:** the one x-intercept, at about $x \approx -1.5$.
- $f'(x) = 0$**:** where the tangent is horizontal — the flat stretch near $x \approx 0.5$ (height -1), the **low point** at $x \approx 4.5$ (height -5), and the **high point** at $x \approx 7$ (height -1).
- **Not continuous:** $x = -3$ and $x = 6$.
- $f'(x)$ **negative:** where the graph falls — roughly $(-7, 0.5)$, $(1, 4.5)$ and $(7, 8)$.
- **Not differentiable:** $x = -3$ and $x = 6$. A function cannot be differentiable where it is not even continuous.
- **Inflection point(s):** where the bending reverses — about $x \approx -1.5$ and $x \approx 2.5$.
- $\lim_{x \to -2} f(x) \approx 1.5$. f is continuous there, so the limit is just the height of the graph.
- $\lim_{x \to 6^-} f(x) = -3$, the *open* circle. The solid dot at -2 is $f(6)$, which is a different question — this is a jump, so the two-sided limit does not exist.

Note $x = -3$ is a **removable** discontinuity: the limit there is 3 (open circle) while $f(-3) = 4$ (solid dot).

(10 pts) **24.** Find the absolute maximum and minimum values of $f(x) = e^x \cos(x)$ on the interval $[0, 2\pi]$. Do NOT use decimal approximations.

Solution:

f is continuous on a closed interval, so the Extreme Value Theorem applies: compare critical values and endpoints.

$$f'(x) = e^x \cos x - e^x \sin x = e^x (\cos x - \sin x)$$

e^x is never 0, so $f'(x) = 0$ when $\cos x = \sin x$, i.e. $\tan x = 1$:

$$x = \frac{\pi}{4}, \qquad x = \frac{5\pi}{4} \quad \text{(both in } [0, 2\pi])$$

Evaluate at the two critical numbers and the two endpoints:

$$f(0) = 1, \qquad f\left(\tfrac{\pi}{4}\right) = \frac{\sqrt{2}}{2} e^{\pi/4}, \qquad f\left(\tfrac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2} e^{5\pi/4}, \qquad f(2\pi) = e^{2\pi}$$

$e^{2\pi}$ is far the largest and $-\dfrac{\sqrt{2}}{2} e^{5\pi/4}$ far the most negative, so

$$\textbf{absolute max } \mathbf{e^{2\pi}} \text{ at } x = 2\pi, \qquad \textbf{absolute min } \mathbf{-\tfrac{\sqrt{2}}{2} e^{5\pi/4}} \text{ at } x = \tfrac{5\pi}{4}$$

THIS IS THE LAST PAGE OF THE EXAM.